

1. Introduction

The Task Management module is designed to handle the creation, assignment, and tracking of individual tasks within projects. It allows teams to break down projects into actionable work items, manage dependencies, track progress, and ensure accountability through detailed task-level controls.

2. Objectives

- Enable teams to efficiently manage tasks within projects.
- Ensure clear visibility and tracking of task progress.
- Improve team collaboration and communication.
- Provide tools to manage task dependencies and priorities.

3. Requirements

ID	Requirement	Description	Priority
FR-3.1	Create/assign/update tasks	Users can create detailed tasks with descriptions, assign them to team members, and update progress. Supports bulk task creation and quick-add functionality for efficiency.	Critical
FR-3.2	Task dependencies	Define relationships between tasks (e.g., Task B cannot start until Task A is complete). Automatically adjusts timelines when dependent tasks are delayed, preventing scheduling conflicts.	High
FR-3.3	Priority levels	Categorize tasks by urgency and importance to help team members focus on what matters most. Visual indicators (colors, icons) make priorities immediately apparent.	Critical

FR-3.4	Status workflow	Track task progression through defined stages. "Blocked" status triggers notifications to project managers, enabling quick issue resolution.	Critical
FR-3.5	Subtasks and checklists	Break down complex tasks into smaller, manageable steps. Completion percentage automatically updates based on subtask progress, providing granular visibility.	Medium
FR-3.6	Time tracking	Log actual hours spent on tasks and compare against estimates. Helps improve future estimations and provides data for resource planning and billing.	High
FR-3.7	File attachments	Attach relevant documents, images, or files directly to tasks. Supports version control to track document changes and maintains context within the task.	High
FR-3.8	Comments and @mentions	Enable threaded discussions within tasks and notify specific team members using @mentions. Keeps all communication contextual and eliminates scattered email threads.	Critical

4. Functional Requirements

4.1 Create/assign/update tasks (FR-3.1)

- Users should be able to create and detail tasks with descriptions.
- Tasks can be assigned to one or multiple team members.
- Progress on tasks can be updated and tracked.
- Bulk task creation and quick-add functionality should be supported for efficiency.

4.2 Task dependencies (FR-3.2)

- Users can define dependencies between tasks.
- System will automatically adjust timelines when dependent tasks are delayed.
- Scheduling conflicts should be prevented and visual indicators provided.

4.3 Priority levels (FR-3.3)

- Tasks should be categorized by urgency: Critical, High, Medium, Low.
- Visual indicators (colors, icons) should help team members identify priorities.

4.4 Status workflow (FR-3.4)

- Track task progression through statuses: Not Started, In Progress, Review, Completed, Blocked.
- "Blocked" status should trigger notifications to project managers for quick issue resolution.

4.5 Subtasks and checklists (FR-3.5)

- Users can create subtasks within larger tasks.
- Completion percentage should automatically update based on subtask progress.

4.6 Time tracking (FR-3.6)

- Users can log actual hours spent on tasks.
- Compare logged data against estimates for better future estimations.
- Provides data useful for resource planning and billing.

4.7 File attachments (FR-3.7)

- Users can attach relevant documents, images, or files to tasks.
- Attachment limit up to 100MB.
- Supports version control to track document changes.

4.8 Comments and @mentions (FR-3.8)

- Enable threaded discussions within tasks.
- Notify specific team members using @mentions.
- Keep all communication contextual and eliminate scattered email threads.

5. Non-Functional Requirements

- The system should be highly available and responsive.
- Ensure data security and confidentiality for sensitive project information.
- The user interface should be intuitive and easy to navigate.
- System should support integration with third-party project management and communication tools.

6. Constraints

- File attachment size is limited to 100MB.
- Users must have appropriate permissions to create, assign, and update tasks.

This document outlines the functional and non-functional requirements necessary to develop a comprehensive Task Management module. By adhering to these requirements, the module aims to provide a robust framework for efficient task management within projects.